

Feature Scaling

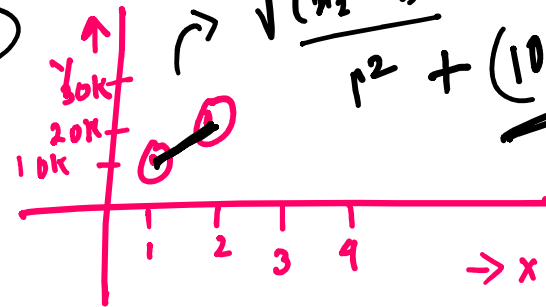
↳ Technique which is used to scale the numerical values within a certain range.

why?

	<u>EXP</u>	<u>SALARY</u>
A	1	10000
B	2	20000
C	3	30000
D	4	40000

✓ Gradient Descent

(1, 10000)
(2, 20000)



$$\sqrt{\frac{(x_2 - x_1)^2 + (y_2 - y_1)^2}{1^2 + (10000)^2}}$$

Two approaches

Standardization

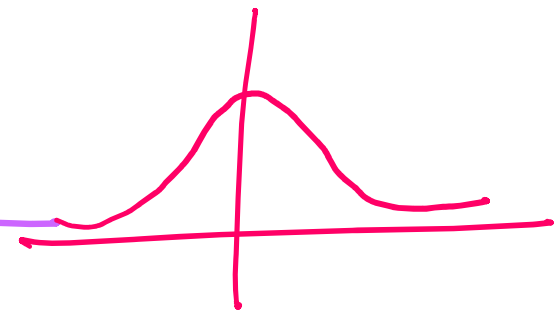
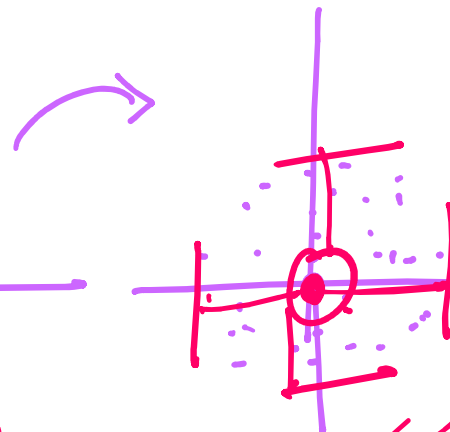
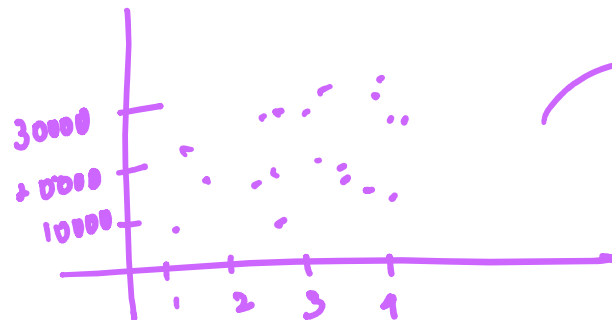
(Z-score Normalization)

Normalization.

1) Standardization :- ✓ Transforms your data in such a way that it has a (mean of 0) and SD = 1

1) Standardization ✓

that it has a mean of 0 and SD = 1
 ↳ spread.



$$x' = \frac{x_i - \mu}{\sigma}$$

original → x_i → mean μ
 $\sigma \rightarrow$ SD

Salary	exp	
10000	1	x_1
20000	2	x_2
30000	3	x_3
40000	4	x_4

$$\text{mean} = \frac{1+2+3+4}{4}$$

$$= 2.5$$

$$SD = 1.11$$

$$x_1 \Rightarrow \frac{1 - 2.5}{1.11} \checkmark$$

$$x_2 \Rightarrow \frac{2 - 2.5}{1.11} \checkmark$$

Workflow

500 rows

1) X_{train} , X_{test} , Y_{train} , Y_{test} .

2) from sklearn.preprocessing import StandardScaler.

Fitting and Transform

③ $\Rightarrow$ Scaler = StandardScaler()

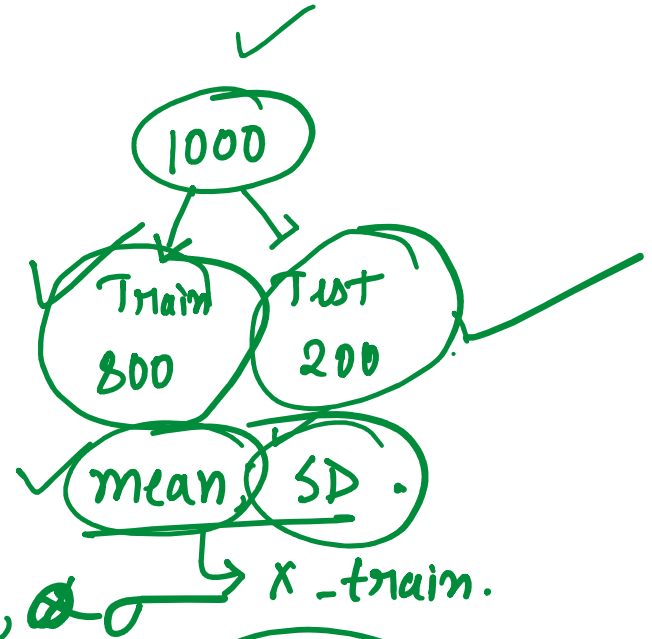
④ $\Rightarrow$ Scaler.fit(X_{train})

mean SD

$\Rightarrow$ $X_{train_scaled} = \text{Scaler.}$

transform(X_{train})

$\Rightarrow$ $X_{test_scaled} = \text{Scaler. transform}$
 (X_{test})



II Normalization (min-max scaling)

→ scales all features so that they strictly fall between a range of 0 - 1
-1 to +1

$$x_{\text{new}} = \frac{x_i - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}$$

<u>Age</u>	<u>Salary</u>
20 ✓	25000
21	26000
22	28000
23	29000
24	30000

$$x_{\text{min}} = 20$$

$$x_{\text{max}} = 24$$

$$x_1 = \frac{20 - 20}{24 - 20} = 0$$

$$x_2 = 21 - 21$$

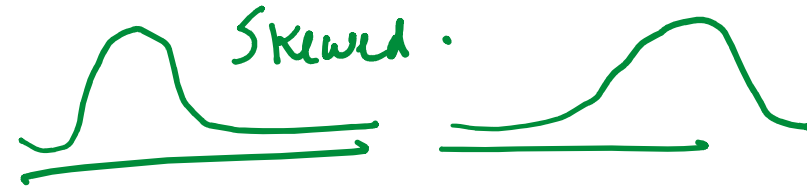
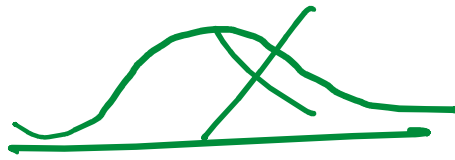
24

30000

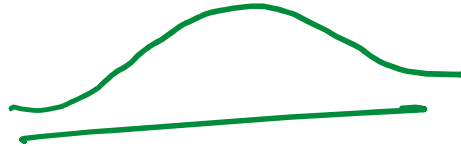
$$x_2 = \frac{20 - 21}{24 - 20}$$

Normalization and Standardization

Normalization -



Standardization -



from sklearn.preprocessing import MinMaxScaler.

Scaler = MinMaxScaler()

Scaler.fit(X_train)

$X_{train}^{scaled} = \text{Scaler.transform}(X_{train})$
 X_{test}

x_{min} x_{max}